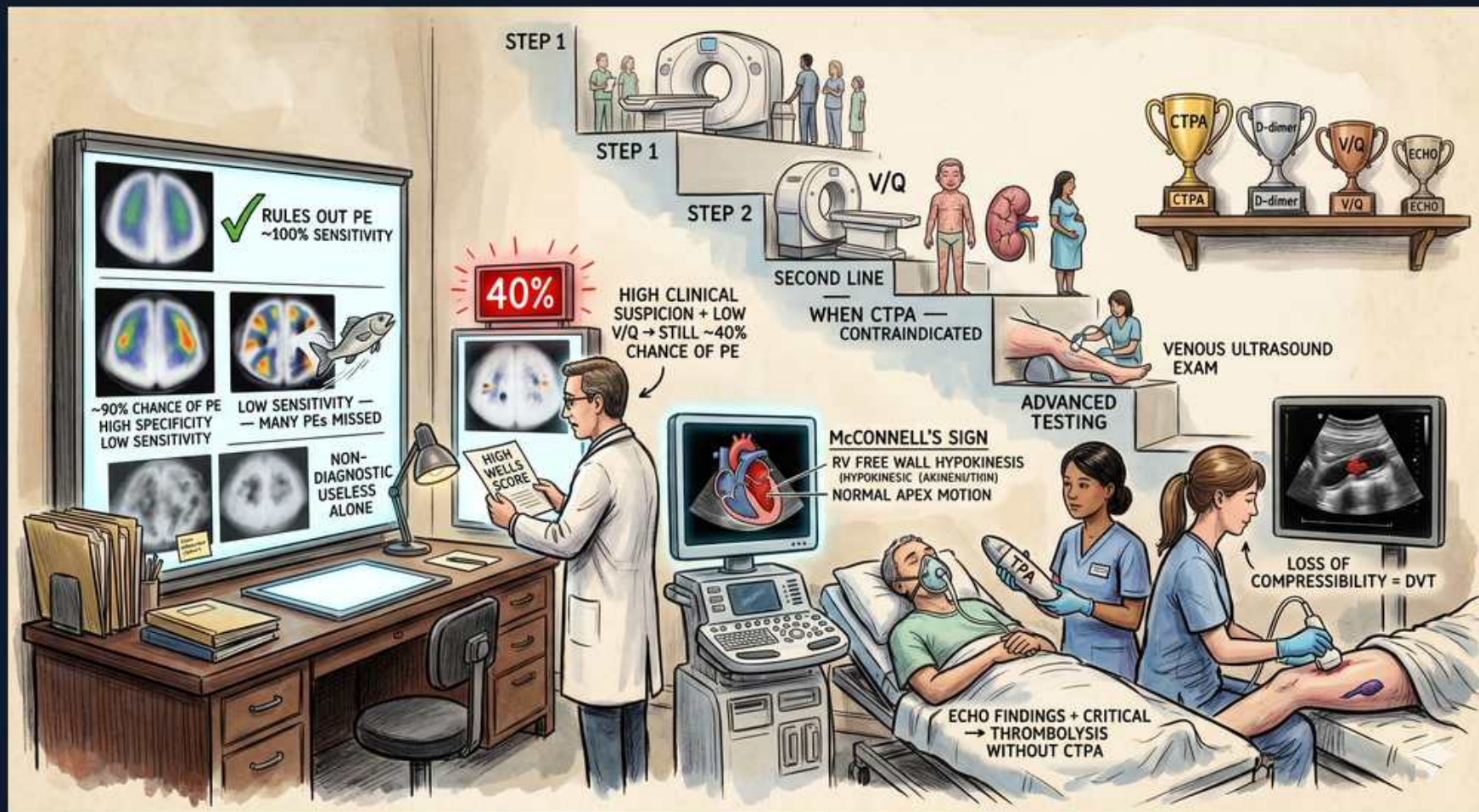


VTE — PE Diagnosis Step 2: V/Q, Echo (McConnell) & Leg Ultrasound



1. V/Q scan — 100% sensitivity rule-out

WHAT YOU SEE

V/Q scan images, '☐ rules out PE 100% sensitivity'

WHAT IT ENCODES

A normal V/Q scan essentially excludes PE (very high negative predictive value). V/Q is the preferred imaging when CTPA is contraindicated (contrast allergy, renal failure, pregnancy).

BOARD TRAP

A NORMAL V/Q rules out PE; a non-diagnostic/intermediate-probability V/Q does not — and it's the common result, demanding further testing.

2. Low specificity — many false PE

WHAT YOU SEE

'90% chance of PE high specificity' vs 'low sensitivity many PE missed'

WHAT IT ENCODES

V/Q probability categories (high/intermediate/low/normal) suffer poor specificity in the middle bands; a 'high-probability' scan in a high-pretest patient confirms PE, but intermediate scans are frequently non-diagnostic.

BOARD TRAP

An intermediate-probability V/Q is NON-DIAGNOSTIC, not reassuring — pair it with pretest probability and proceed to CTPA or ultrasound.

3. Non-diagnostic alone

WHAT YOU SEE

'Non-diagnostic — useless alone'

WHAT IT ENCODES

V/Q must be interpreted with clinical pretest probability; alone, many scans are indeterminate. Abnormal underlying lung (COPD, infiltrates) degrades its value, favoring CTPA when feasible.

BOARD TRAP

Don't act on a V/Q in isolation — combine with Wells; baseline lung disease makes V/Q unreliable, so CTPA is preferred there.

4. 40% PE despite low V/Q

WHAT YOU SEE

Red '40%' sign, clinician reading

WHAT IT ENCODES

High clinical suspicion plus a low-probability V/Q still leaves a meaningful (~40%) residual chance of PE — pretest probability overrides a discordant scan.

BOARD TRAP

High pretest probability + low-probability V/Q ≠ excluded — clinical gestalt wins; escalate imaging.

5. CTPA contraindicated → V/Q

WHAT YOU SEE

Step 1/Step 2 path to V/Q 'when CTPA contraindicated'

WHAT IT ENCODES

Diagnostic sequencing: CTPA is first-line; V/Q is the second-line/alternative when CTPA is contraindicated. The stepwise approach conserves contrast and radiation appropriately.

BOARD TRAP

V/Q is the go-to ALTERNATIVE when contrast/renal/pregnancy preclude CTPA — not an inferior routine test for everyone.

6. Trophy shelf — test ranking

WHAT YOU SEE

Trophies: CTPA, V/Q, DVT ultrasound

WHAT IT ENCODES

Hierarchy of confirmation: CTPA (gold standard) > V/Q (when CTPA contraindicated) > venous ultrasound (confirms DVT, indirectly supports PE).

BOARD TRAP

A positive leg ultrasound in a symptomatic patient can confirm VTE and justify treatment WITHOUT chest imaging — useful when CTPA/V/Q are unavailable.

7. McConnell's sign

WHAT YOU SEE

Echo of dilated RV, 'McConnell's sign'

WHAT IT ENCODES

McConnell's sign on echo = RV free-wall akinesia/hypokinesia with PRESERVED apical motion — relatively specific for acute PE. Echo also shows RV dilation, septal flattening, and elevated PASP.

BOARD TRAP

McConnell's sign = free-wall akinesis with SPARED apex (specific for acute PE) — distinguishes PE strain from global RV failure.

8. Echo critical without CTPA

WHAT YOU SEE

Bedside echo, 'echo findings critical — thrombolysis without CTPA'

WHAT IT ENCODES

In an unstable, hypotensive patient too sick for CTPA, bedside echo showing acute RV strain/McConnell's supports empiric thrombolysis for presumed massive PE.

BOARD TRAP

For a crashing patient, RV strain on bedside echo can justify EMPIRIC thrombolysis — don't delay reperfusion to transport for CTPA.

9. Venous ultrasound exam

WHAT YOU SEE

'Venous ultrasound exam / advanced testing'

WHAT IT ENCODES

Compression ultrasound is the first-line test for suspected DVT — safe, no contrast, bedside. A positive proximal study in a PE-suspect patient confirms VTE and warrants treatment.

BOARD TRAP

Whole-leg vs proximal compression matters: a negative single proximal scan with high suspicion warrants a repeat in ~1 week or whole-leg imaging.

10. Loss of compressibility = DVT

WHAT YOU SEE

Ultrasound clot, 'loss of compressibility = DVT'

WHAT IT ENCODES

The diagnostic ultrasound finding for DVT is a non-compressible vein under probe pressure; absent compressibility = intraluminal thrombus. Color Doppler supports but compression is the key sign.

BOARD TRAP

Loss of COMPRESSIBILITY (not just absent Doppler flow) defines DVT on ultrasound — flow artifacts can mislead; compression is the standard.

11. Step 1 → Step 2 sequence

WHAT YOU SEE

'Step 1' / 'Step 2' workflow arrows

WHAT IT ENCODES

Diagnostic workflow: confirm pretest probability and attempt definitive imaging (CTPA), then escalate to second-line (V/Q) or advanced testing (echo, leg ultrasound) based on contraindications and stability.

BOARD TRAP

The pathway is conditional — choose the next test by contraindications and hemodynamics, not a fixed order for every patient.

12. Non-diagnostic V/Q → escalate

WHAT YOU SEE

'Non-diagnostic useless alone' panel

WHAT IT ENCODES

When V/Q is indeterminate, escalate to CTPA (if now feasible) or compression ultrasound; a positive leg study closes the loop for VTE treatment.

BOARD TRAP

An indeterminate V/Q is an indication to ESCALATE, not to stop — pairing with leg ultrasound often resolves the case.
